GEOMETRY REFERENCE SHEET

✓ (points)

$\mathbb{E}$ = euclidean space (affine space of dim ≥ 3) $D(\mathbb{A})$: $\mathbb{V}$ attached to $\mathbb{A}$. ; $\dim(D(\mathbb{A})) = \dim(\mathbb{A})$. direction space.

$\mathbb{E}^m$ = m-dimensional euclidean space (euclidean properties)

$\mathbb{V}^m$ = space of vectors in $\mathbb{E}^m$; $\mathbb{V} = \{\vec{AB}$, where A, B points in $\mathbb{E}\} = \mathbb{E} \times \mathbb{E}$ (fixed O) (vectors)

$\mathbb{A}$ = affine space = set of points to which we attach $\mathbb{V}$ and 2 maps. < 2 points => vector; vector + point => point. (all points initially equal)

$\mathbb{A}^m$ = m-dimensional affine space (act on points)

✓✓ **ORIENTED SEGMENT**

$[AB]$; $A \to B$.

✓✓ **EQUIPOLENCY**: (equiv. rel.)

$[AB] \sim [CD] \Leftrightarrow$ same midpoints

✓✓ **FREE VECTOR**

$B - A = \vec{AB}$

✓✓ **LENGTH OF A VECTOR**

$|\vec{AB}| = |B-A| = \sqrt{x^2+y^2+\dots}$

✓✓ **AFFINE SUBSPACE**

$$S \subseteq \mathbb{A} \Leftrightarrow \begin{cases} \exists P \\ \exists U \leq D(\mathbb{A}) \end{cases}$$

s.t. $S = U + P = \{v + P, v \in U\}$

S = point $\Leftrightarrow \dim U = 0$.

S = line $\Leftrightarrow \dim U = 1$

S = plane $\Leftrightarrow \dim U = 2$

S = hyperplane $\Leftrightarrow \dim U = \dim \mathbb{A}^m - 1$.

✓✓ **THEOREM 1.10.**

$K = (O, \mathcal{B})$; $K' = (O, \mathcal{B}') \to \mathbb{A}^m$.

$P \in \mathbb{A}^m$.

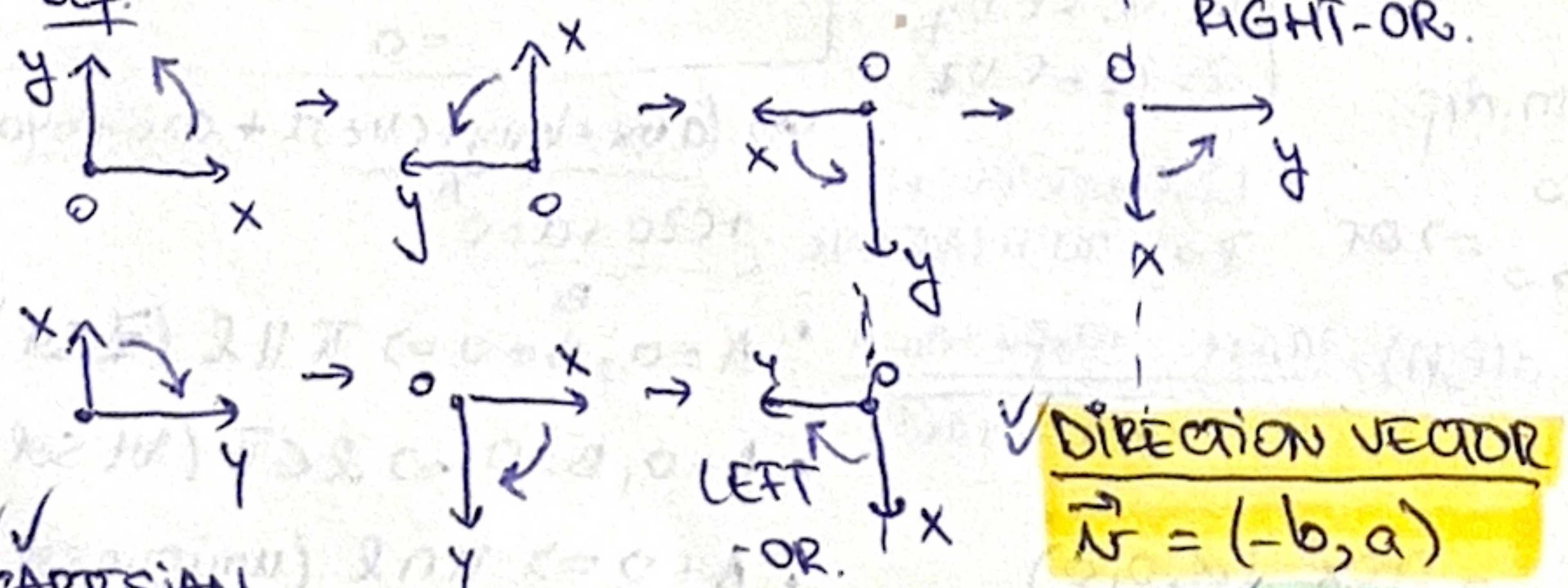
$$[P]_{K'} = M_{B'B}([P]_K - [O']_K)$$
$$= M_{BB'}^{-1}([P]_K - [O']_K)$$
$$= M_{B'B} \cdot [P]_K + [O']_{K'}$$

✓✓ **LEFT/ RIGHT - ORIENTATION**

$\mathcal{E} = (e_1, e_2)$, $\mathcal{F} = (f_1, f_2) \in \mathbb{V}^2$.

- $\det(M_{\mathcal{E}\mathcal{F}}) > 0 \Rightarrow$ same orientation
- $\det(M_{\mathcal{E}\mathcal{F}}) < 0 \Rightarrow \neq$ orientation.

Def:



✓✓ **CARTESIAN ~~LINEAR~~ EQUATION of a line ($\mathbb{A}^2$)**

$ax + by + c = 0$.

$D(\ell)$: $ax + by = 0$ (dir. space) $\Big|$ $\alpha = -\dfrac{c}{a}$, $\beta = -\dfrac{c}{b}$

✓✓ **EQ. of line w intersection:** $\dfrac{x}{\alpha} + \dfrac{y}{\beta} = 1$. $\to \cap Ox : (\alpha, 0)$; $\to \cap Oy : (0, \beta)$

✓✓ **VECTOR SPACE**

$\mathbb{V}$ with
- vector addition
- scalar multiplication

✓ **FRAME $\mathbb{E}^2$**: $K = (O, \mathcal{B})$

$\vec{OP} = x_P \cdot \vec{i} + y_P \cdot \vec{j}$

$P(x_P, y_P) \Leftrightarrow \vec{OP}(x_P, y_P)$

$P = A + t \cdot \vec{v}$

$(x, y) = (x_A, y_A) + t \cdot (a, b)$

$[P]_K = \begin{bmatrix} x_P \\ y_P \end{bmatrix} \in \mathbb{R}^2$ (point)

$[a]_B = \begin{bmatrix} a_x \\ a_y \end{bmatrix} \in \mathbb{R}^2$ (vector)

✓✓ **FRAME $\mathbb{A}^m$**: $K = (O, \mathcal{B})$.

$\vec{OP} = x_1 i_1 + \dots + x_m i_m$

$[P]_K = \begin{bmatrix} x_1 \\ \vdots \\ x_m \end{bmatrix} \in \mathbb{R}^m$

$[a]_B = \begin{bmatrix} a_1 \\ \vdots \\ a_m \end{bmatrix} \in \mathbb{R}^m$

✓ **LINES IN $\mathbb{A}^2$:**

$S = \underset{\text{vector space}}{U} + \underset{\text{point}}{P}$

✓ **P, Q on line S:**

$\vec{QP} = t \cdot v$ direction vector of S. $\begin{cases} Q \to \text{fixed on line } S. \\ t \in \mathbb{R} \\ v \in U = D(S). \end{cases}$

t: unique scalar $\in \mathbb{R}$

$(\forall) O \in \mathbb{A}^2$: $\boxed{\vec{OP} = \vec{OQ} + \vec{QP} = \vec{OQ} + t \cdot v}$ VECTOR EQUATION OF THE LINE S RELATIVE TO O.

" $\in \mathbb{A}^3$: ↗

✓ **DIRECTION VECTOR**

$\vec{v} = (-b, a)$

✓ **NORMAL VECTOR**

$n = (a, b, c)$

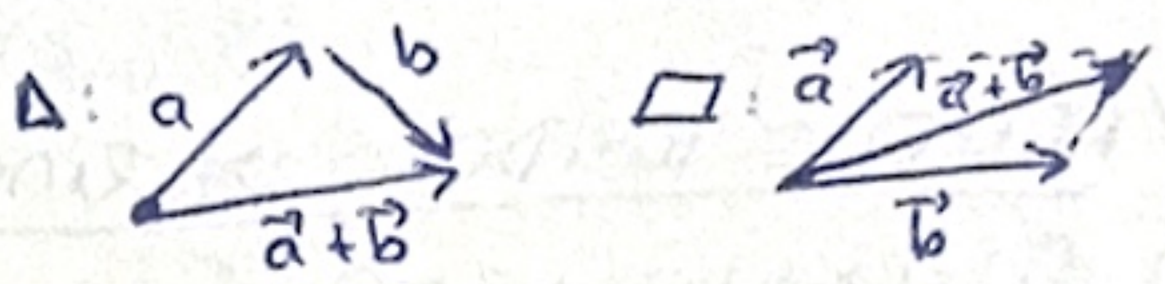
PROJECT A POINT ON LINE ℓ: (ℓ)

$A \in \ell$; $v \to$ dir. of line ℓ

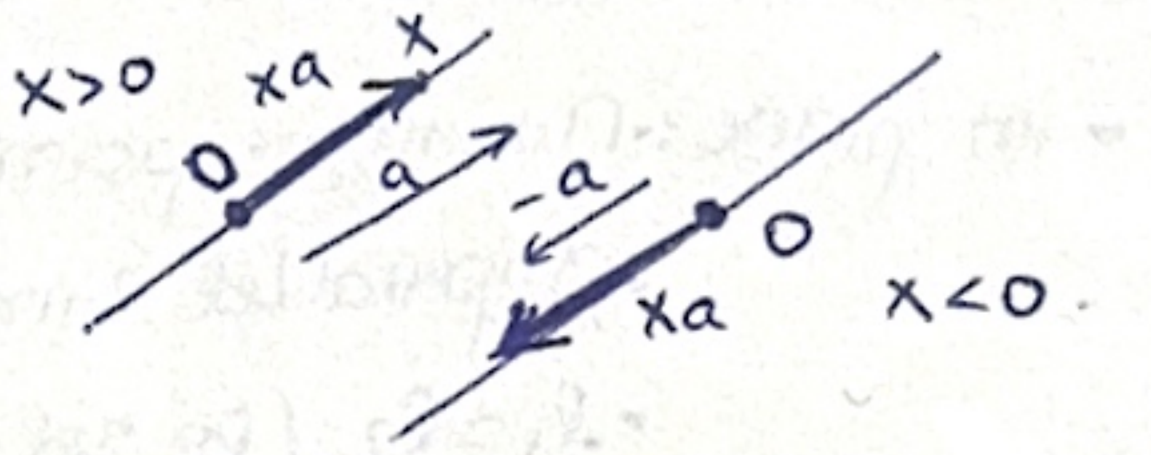
$\vec{AP'} = Pr_v^{\perp}(\vec{AP}) = \dfrac{\langle AP, v \rangle}{|v|^2} v$

$P' = A + \vec{AP'}$ (projected point)

✓✓ **VECTOR ADDITION**



✓✓ **SCALAR MULTIPLICATION**



$$\underset{\text{scalar}}{x} \cdot \underset{\text{vector}}{a} = \begin{cases} \vec{Ox}, & a \neq \vec{0}, x > 0 \\ -(|x| a), & a \neq \vec{0}, x < 0 \\ \vec{0}, & a = \vec{0} / x = 0. \end{cases}$$

?✓ **CARTESIAN EQ. w determinant:**

$\begin{vmatrix} x - x_Q & y - y_Q \\ v_x & v_y \end{vmatrix} = 0 \Rightarrow v, \vec{QP}$ lin. dep.

↓ half-planes described by line S.

$| \; | < 0$ ⟵ ⟶ $| \; | > 0$

DIST. FROM P to line/plane = MAGNITUDE of PP'.
$P'' = 2P' - P$ (symm. point)

?✓ **DIRECTION VECTOR:**

v direction vector for $S \Leftrightarrow v$ lin. dep. on $\vec{QP}$

PROJECT POINT ON PLANE
- P point to be proj.
- $n(a, b, c)$
- $\vec{u}$.

$\delta = \dfrac{a x_0 + b y_0 + c z_0 + d}{|n|}$

$P' = P - \delta n$

✓✓ **EQ. OF LINE w FIXED FRAME.**

$$(S): \begin{cases} x = x_Q + t \cdot v_x \\ y = y_Q + t \cdot v_y \end{cases}$$
PARAMETRIC EQUATIONS

$z = z_Q + t \cdot v_z$ ($\mathbb{A}^3$)

✓✓ **MATRIX FORM OF A LINE w FIXED FRAME**

$S: \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} x_Q \\ y_Q \end{bmatrix} + t \cdot \begin{bmatrix} v_x \\ v_y \end{bmatrix}$ $\mathbb{A}^2$

✓✓ **SYMMETRIC ~~CARTESIAN~~ EQUATIONS OF line S**

$\dfrac{x - x_Q}{v_x} = \dfrac{y - y_Q}{v_y} = t$

$v(v_x, v_y)$ dir. vector;

$\vec{n} \perp$ plane

✓ $\mathbb{A}^m$:

$$S: \begin{bmatrix}x_1\\ \vdots \\ x_m\end{bmatrix} = \begin{bmatrix} q_1 \\ \vdots \\ q_m\end{bmatrix} + t_1\begin{bmatrix} v_{1,1}\\ \vdots\\ v_{1,m}\end{bmatrix} + \dots + t_d \cdot \begin{bmatrix} v_{d,1}\\ \vdots \\ v_{d,m}\end{bmatrix}$$

$S = \{P\in \mathbb{A}^m : \vec{OP} = \vec{OQ} + t_1 \cdot v_1 + \dots + t_d \cdot v_d\}$

$\mathbb{U} = \langle v_1,\dots,v_d\rangle$ basis. $\le \mathbb{V}$.

$\dim \mathbb{U} = d < m$.

✓ Cartesian form of a plane

$\pi: ax+by+cz+d=0$.

✓ parametric form of a plane

$(x,y,z) = P_0 + s\cdot\vec{u} + t\cdot\vec{v}$

?✓ RELATIVE POSITIONS of lines

$\begin{cases} l_1: a_1x+b_1y+c_1=0 \\ l_2: a_2x+b_2y+c_2=0\end{cases} \to \cap?$

- in plane: $\cap$ ⇒ unique point. (sol.)
  - $\not\cap$ ⇒ parallel lines (∄ sol.)
  - $l_1 = l_2$ (∞ sol.)

✓ VECTOR EQUATION OF the PLANE "S"

$\vec{OP} = \vec{OQ} + sv + tw$.

v, w dir. vectors

Q base point.

(∀) $O \in \mathbb{A}^3$.

✓ DISTANCE BETWEEN TWO ∥ PLANES

$\begin{cases} ax+by+cz+d_1=0 \\ ax+by+cz+d_2=0\end{cases}$

$d = \dfrac{|d_1-d_2|}{\sqrt{a^2+b^2+c^2}}$

✓ PARAMETRIC EQUATIONS OF the PLANE "S"

(fixed frame); s, t parameters

$$S: \begin{cases} x = x_Q + s\cdot v_x + t\cdot w_x \\ y = y_Q + s\cdot v_y + t\cdot w_y \\ z = z_Q + s\cdot v_z + t\cdot w_z \end{cases}$$

✓ MATRIX FORM:

$$S: \begin{bmatrix} x \\ y \\ z\end{bmatrix} = \begin{bmatrix} x_Q \\ y_Q \\ z_Q \end{bmatrix} + s\cdot\begin{bmatrix} v_x \\ v_y \\ v_z\end{bmatrix} + t\cdot\begin{bmatrix} w_x \\ w_y \\ w_z\end{bmatrix}$$

$\Delta' = \Delta - \dfrac{\langle AD, n\rangle}{|n|^2} \cdot n$ (plane)

$\Delta' = A + \dfrac{\langle AD, AB\rangle}{|AB|^2} AB$ (line)

?✓ $P(x,y,z) \in$ plane $S \Leftrightarrow \vec{QP}, v, w$ lin. dep.

$$\Leftrightarrow \begin{vmatrix} x-x_Q & y-y_Q & z-z_Q \\ v_x & v_y & v_z \\ w_x & w_y & w_z \end{vmatrix} = 0.$$

✓ COPLANAR POINTS:

Q, E, F, G coplanar $\Leftrightarrow \vec{QE}, \vec{QF}, \vec{QG}$ lin. dep.

$$\Leftrightarrow \begin{vmatrix} x_E-x_Q & y_E-y_Q & z_E-z_Q \\ x_F-x_Q & y_F-y_Q & z_F-z_Q \\ x_G-x_Q & y_G-y_Q & z_G-z_Q \end{vmatrix} = 0.$$

$\nearrow <0$, $\searrow >0$ ⇒ or.

✓ PLANE in $\mathbb{A}^3$:

$ax+by+cz+d=0$

↓ $K=(O,B)$

$D(\pi): ax+by+cz=0$

$\alpha = -\dfrac{d}{a} \to \cap OX: (\alpha,0,0)$

$\beta = -\dfrac{d}{b} \to \cap OY: (0,\beta,0)$

$\gamma = -\dfrac{d}{c} \to \cap OZ: (0,0,\gamma)$

✓✓ HYPERPLANES = $\mathbb{A}^m$ with dim $m-1$.

$H \to$ hyperplane. $\boxed{a_1x_1+\dots+a_mx_m=b}$

$(v_1,\dots,v_{m-1})$ basis of $D(H)$.

$K = (O, e_1, \dots, e_m)$ frame of $\mathbb{A}^m$

$\vec{n} = (a_1, \dots, a_n)$

$\vec{n} \perp$ hyperpl.

✓✓ PARAMETRIC EQUATIONS of H:

$$H: \begin{cases} x_1 = q_1 + t_1 v_{1,1} + \dots + t_{m-1}\cdot v_{m-1,1} \\ \vdots \\ x_m = q_m + t_1\cdot v_{1,m} + \dots + t_{m-1}\cdot v_{m-1,m}\end{cases}$$

$$H: \begin{bmatrix} x_1 \\ \vdots \\ x_m\end{bmatrix} = \begin{bmatrix} q_1 \\ \vdots \\ q_m\end{bmatrix} + t_1\cdot\begin{bmatrix} v_{1,1} \\ \vdots \\ v_{1,m}\end{bmatrix} + \dots + t_{m-1}\cdot\begin{bmatrix} v_{m-1,1} \\ \vdots \\ v_{m-1,m}\end{bmatrix}$$

$P \in H \Leftrightarrow \vec{n}\cdot\vec{QP} = 0$

$D(H): a_1x_1 + \dots + a_nx_n = 0$

?✓ RELATIVE POSITIONS of planes

$\begin{cases} \pi_1: a_1x+b_1y+c_1z+d_1=0 \\ \pi_2: a_2x+b_2y+c_2z+d_2=0\end{cases} \to \cap?$

- $\cap$ ⇒ line (sol.) (if rank M = rank $\tilde{M}$ = 2)
- $\not\cap$ ⇒ parallel planes (∄ sol) (if rank M < rank $\tilde{M}$)
- $\pi_1 = \pi_2$ (sol. depends on 2 params) (if rank M = rank $\tilde{M}$ = 1)

? point-normal equation $n(a,b,c)$

$a(x-x_0)+b(y-y_0)+c(z-z_0)=0$

✓ LINE in $\mathbb{A}^3$:

$l: \begin{cases} a_1x+b_1y+c_1z+d_1=0 \\ a_2x+b_2y+c_2z+d_2=0\end{cases} \to D(l): \begin{cases} a_1x+b_1y+c_1z=0 \\ a_2x+b_2y+c_2z=0\end{cases}$

?✓ RELATIVE POSITIONS of TWO LINES in $\mathbb{A}^3$:

$l_1: \begin{cases} a_1x+b_1y+c_1z+d_1=0 \\ a_2x+b_2y+c_2z+d_2=0\end{cases}, \quad l_2: \begin{cases} a_3x+b_3y+c_3z+d_3=0 \\ a_4x+b_4y+c_4z+d_4=0\end{cases}$

↓ parametric equations:

$l_1: \begin{cases} x = x_1 + t\cdot v_x \\ y = y_1 + t\cdot v_y \\ z = z_1 + t\cdot v_z\end{cases} \quad l_2: \begin{cases} x = x_2 + t\cdot u_x \\ y = y_2 + t\cdot u_y \\ z = z_2 + t\cdot u_z\end{cases}$

- $u, v \to$ proportional ⇒ $l_1 \parallel l_2$.
- $l_1 \parallel l_2$ and $P \in l_1, P \in l_2 \Rightarrow l_1 = l_2$.
- $l_1 \nparallel l_2 \to l_1, l_2$ coplanar. if: $\begin{vmatrix} x_1-x_2 & y_1-y_2 & z_1-z_2 \\ v_x & v_y & v_z \\ u_x & u_y & u_z\end{vmatrix} = 0 \Rightarrow \cap$.
- $\left.\begin{matrix} l_1 \nparallel l_2 \\ l_1 \not\cap l_2\end{matrix}\right\} \Rightarrow l_1, l_2$ skew.

?✓ RELATIVE POSITIONS of LINE and PLANE. ($\mathbb{A}^3$)

$\pi: ax+by+cz+d=0$;

$l: \begin{cases} x = x_0 + t\cdot v_x \\ y = y_0 + t\cdot v_y \\ z = z_0 + t\cdot v_z\end{cases}$

- $\cap$? which points $\in l$ satisfy

$\boxed{a(x_0+t\cdot v_x) + b(y_0+t\cdot v_y) + c\cdot(z_0+t\cdot v_z) = 0}$

$\Leftrightarrow \underbrace{(av_x+bv_y+cv_z)}_{A}t + \underbrace{ax_0+by_0+cz_0+d}_{B}=0$

✓ DISTANCE FROM POINT TO HYPERPLANE

$d(P,H) = \dfrac{|a_1p_1+\dots+a_np_n+a_{n-1}|}{\sqrt{a_1^2+\dots+a_n^2}}$

- $A=0, B\neq0 \Rightarrow \pi \parallel l$ (∄ sol.)
- $A=0, B=0 \Rightarrow l \in \pi$ (∀t sol.)
- $A\neq0 \Rightarrow \pi \cap l$ (unique sol.)

$t_0 = -\dfrac{ax_0+by_0+cz_0+d}{av_x+bv_y+cv_z}$

? P belongs to H (hyperplane)

$$\begin{vmatrix} x_1-q_1 & x_2-q_2 & \dots & x_m-q_m \\ v_{1,1} & v_{1,2} & \dots & v_{1,m} \\ \vdots & & & \\ v_{m-1,1} & v_{m-1,2} & & v_{m-1,m}\end{vmatrix} = 0$$

✓ VOLUME OF PARALLELIPIPED

$Vol(P) = |\langle a\times b, c\rangle| = \left|\begin{vmatrix} a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{vmatrix}\right|$

(MODULUS!!!) → for A, V.

✓✓ COPLANARITY: $[a,b,c]=0$ $\Updownarrow$ $V_{parallelip}=0$.

~~Lab + DS~~

✓✓ ~~0. try commands in first 3 pages of tutorial~~

BOX PRODUCT ∩ MIXED PRODUCT.

$\langle a\times b, c\rangle = [a,b,c] = [b,c,a] = [c,a,b] = -[b,a,c] = -[a,c,b] = -[c,b,a]$.

✓✓ COORDINATE FORMULA (*)

$d(l,l') = \frac{|a_x(x_B-x_A)+a_y(y_B-y_A)+a_z(z_B-z_A)|}{\sqrt{a_x^2+a_y^2+a_z^2}}$

→ $\boxed{a = v\times w = (a_x, a_y, a_z)}$

✓ PARALLELIPIPED FORMULA

$d(l,l') = \frac{Vol(P)}{Area(F)} = \left|\frac{[v,w,\vec{AB}]}{|v\times w|}\right|$

✓ Gram-Schmidt.

K = (O, B) non-orthonormal frame. → scalar prod. in given reference K → Gram matrix. ?
→ find K' using Gram-Schmidt.

$\boxed{\begin{array}{l}\text{LINE: } ax+by+c=0 \\ \text{PLANE: } ax+by+cz+d=0\end{array}}$ ✓

↓

K' = (O, B')

B = (e₁, …, eₘ)

B' = (e₁', …, eₘ')

1) $e_1' = e_1$

$e_2' = e_2 - \frac{\langle e_1', e_2\rangle}{\langle e_1', e_1'\rangle}\cdot e_1' = e_2 - Pr^{\perp}_{e_1'}(e_2)$

$e_3' = e_3 - \frac{\langle e_1', e_3\rangle}{\langle e_1', e_1'\rangle}\cdot e_1' - \frac{\langle e_2', e_3\rangle}{\langle e_2', e_2'\rangle}\cdot e_2' = e_3 - Pr^{\perp}_{e_1'}(e_3) - Pr^{\perp}_{e_2'}(e_3)$

⋮

2 : $e_1 \frown e_2$

3 : $e_1 \frown e_3$, $e_2 \frown e_3$

2) $B' = \left\{\frac{e_1'}{|e_1'|}, \dots, \frac{e_m'}{|e_m'|}\right\}$. orthonormal basis.

✓ Jacobi identity

$\boxed{(a\times b)\times c + (b\times c)\times a + (c\times a)\times b = 0}$

✓ DOUBLE CROSS FORMULA

$(a\times b)\times c = \langle a,c\rangle b - \langle b,c\rangle a$

$a\times (b\times c) = \langle a,c\rangle b - \langle a,b\rangle c$

✓✓ VECTOR PROJECTION (*)

$d(l,l') = \frac{|\langle v\times w, \vec{AB}\rangle|}{|v\times w|}$

v → dir. vector of l

w → dir. vector of l'

AB → vector that connects l with l'.

v×w → dir. of common ⊥.

? HYPERPLANE:

H: $a_1x_1 + \dots + a_mx_m = b$.

K orthonormal

Q(q₁, …, qₘ) fixed ∈ H.

$b = a_1q_1 + \dots + a_mq_m$.

$\Rightarrow a_1x_1 + \dots + a_mx_m = a_1q_1 + \dots + a_mq_m$

$\Rightarrow a_1(x_1-q_1) + \dots + a_m(x_m-q_m) = 0$.

n(a₁, …, aₘ); P ∈ H ⟺

$\langle n, \vec{QP}\rangle = 0 \iff n \perp \vec{QP}$

⇒ the vector n of coefficients is orthogonal to any vector ∥ H.

⇒ orthogonal to H.

✓✓ $\boxed{d(P,H) = \frac{|a_1p_1 + \dots + a_mp_m - b|}{\sqrt{a_1^2 + \dots + a_m^2}}}$

✓ $\boxed{\begin{array}{l} d(S,H) = d(P,H) \\ S \parallel H, \text{subspace.}\end{array}}$

? LOCI of POINTS: $\mathcal{L}(S,c) = \{P\in \mathbb{E}^m : d(P,S) = c\}$.

(the set of points at distance c from S).

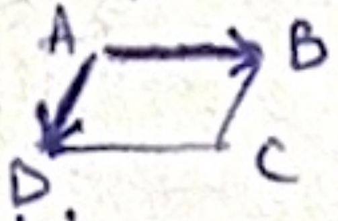
AREA OF POLYGONS:

- S square $\overset{l=1}{\Rightarrow}$ Area(S) = 1
- S₁, S₂ similar polygons w ratio x, Area(S₁) = x²·Area(S₂)
- S₁, S₂ disjoint polygons, Area(S₁ ∪ S₂) = Area(S₁) + Area(S₂)

✓ Area of ΔABC

$\frac{1}{2}|AB\times AC|$ → 2D: $\frac{1}{2}\left|\begin{vmatrix} x_A & y_A & 1 \\ x_B & y_B & 1 \\ x_C & y_C & 1\end{vmatrix}\right|$

✓ AREA OF PARALLELOGRAM



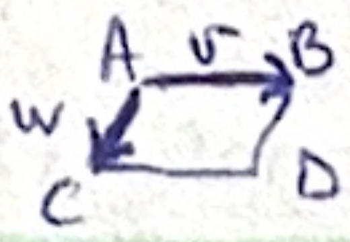
$A(ABCD) = \left|\begin{vmatrix} x_A & y_A & 1 \\ x_B & y_B & 1 \\ x_D & y_D & 1\end{vmatrix}\right| = \left|\begin{vmatrix} v_x & v_y \\ w_x & w_y\end{vmatrix}\right|$

$\parallel$

$|AB\times AC|$

$v(v_x, v_y) = \vec{AB}$

$w(w_x, w_y) = \vec{AD}$



✓✓ $A(ABDC) = [v,w] = [\vec{AB}, \vec{AC}]$ = ORIENTED AREA OF PARALL.

✓ ORIENTED AREA OF Δ: $A_{or}(ABC) = A_{or}(ABCD)/2$.

✓ NORMAL VECTOR OF A HYPERPLANE

n ⊥ H

✓ Hyperplane = line (in $\mathbb{E}^2$)

l : x + 3y − 3 = 0

⇒ n(1, 3) normal vector.

✓ Hyperplane = plane (in $\mathbb{E}^3$)

π : $2x - y + \frac{1}{3}z + 7 = 0$

⇒ n(6, −3, 1) normal vector.

✓✓ ORIENTED VOLUME (Parallelip.)

$V_{or}(P) = [a,b,c]$

✓✓ ORIENTED VOLUME (tetrah.)

$V_{or}(ABCD) = [\vec{AB}, \vec{AC}, \vec{AD}]/6$

✓ ANGLES BETWEEN LINES

l₁, l₂ ∈ $\mathbb{E}^2$.

- v₁, v₂ dir. vectors for those lines.

$\boxed{\cos\sphericalangle(v_1,v_2) = \frac{\langle v_1, v_2\rangle}{|v_1|\cdot|v_2|}}$

- n₁, n₂ normal vectors.

$\cos\sphericalangle(n_1,n_2) = \frac{\langle n_1, n_2\rangle}{|n_1|\cdot|n_2|}$

- v₁ for l₁ and n₂ for l₂

$\boxed{\sphericalangle l_1, l_2 = \frac{\pi}{2} - \arccos\left(\left|\frac{\langle v_1, n_2\rangle}{|v_1|\cdot|n_2|}\right|\right)} \in \left[0, \frac{\pi}{2}\right)$

?✓ COMMON ⊥ OF TWO SKEW LINES

$d = \begin{cases} \pi: \det\begin{pmatrix} x-x_A & y-y_A & z-z_A \\ v_x & v_y & v_z \\ a_x & a_y & a_z\end{pmatrix} = 0 \\ \pi': \det\begin{pmatrix} x-x_B & y-y_B & z-z_B \\ w_x & w_y & w_z \\ a_x & a_y & a_z\end{pmatrix} = 0\end{cases}$

→ l passes through A with direction vector v.

$\boxed{\begin{array}{c} a = v\times w \\ \text{direction of the } \perp\end{array}}$

→ l passes through B with direction vector w.

✓✓ SIGN OF BOX F:

$[a,b,c]\begin{cases} >0 \Rightarrow (a,b,c) \text{ r-or.} \\ =0 \Rightarrow (a,b,c) \text{ not basis.} \\ <0 \Rightarrow (a,b,c) \text{ l-or.}\end{cases}$

GENERALIZATION:

$\mathbb{E}^m$: l₁, l₂ with v₁, v₂

H₁, H₂ with n₁, n₂

1) l₁, l₂ → 2 supplem. angles: $\sphericalangle(v_1,v_2)$, $\sphericalangle(-v_1,v_2)$

$\cos\sphericalangle(v_1,v_2) = \frac{\langle v_1, v_2\rangle}{|v_1|\cdot|v_2|}$

2) H₁, H₂ : $\cos\sphericalangle(n_1, \dots$

3) l₁, H₁ :

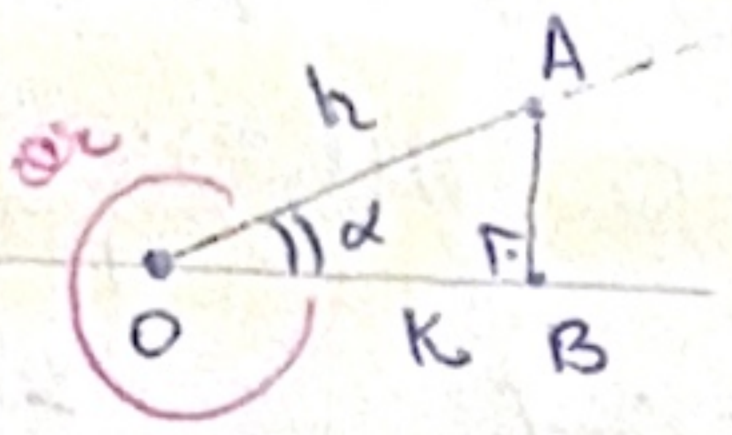
✓ TRIPLE CROSS PRODUCT

$(a\times b)\times(c\times d) = b\cdot[a,c,d] - a\cdot[b,c,d]$
$= c\cdot[a,b,d] - d\cdot[a,b,c]$

✓ SINE, COSINE:

$\sin\alpha = \frac{|AB|}{|OA|}$ $\frac{\text{cat. op}}{\text{ip}}$

$\measuredangle(h,k)$



$\cos\alpha = \begin{cases} 0, & \alpha = 90^\circ \\ \frac{|OB|}{|OA|}, & \alpha < 90^\circ \\ -\frac{|OB|}{|OA|}, & \alpha > 90^\circ \end{cases}$ $\frac{\text{cat. al.}}{\text{ip.}}$

✓ CONGRUENT ANGLES

(⇔) cosines are equal.

✓ UNORIENTED ANGLE

$a = \vec{OA}$, $b = \vec{OB}$

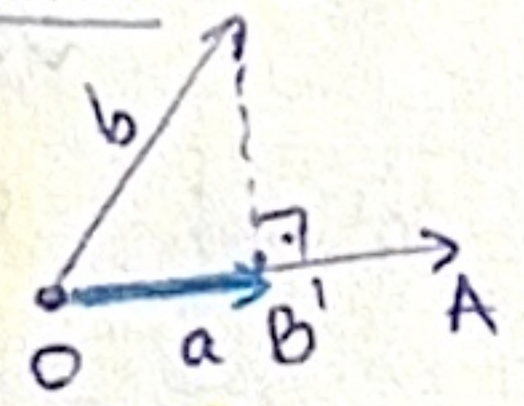
$\measuredangle(a,b) = \measuredangle AOB$.

$\cos\measuredangle(a,b) = 0 \Rightarrow a \perp b$.

✓ ORTHOGONAL PROJECTION (linear) MAPS

$pr^{\perp}_{a} : \mathbb{V} \to \mathbb{R}$ (length)

$Pr^{\perp}_{a} : \mathbb{V} \to \mathbb{V}$ (vector)



$pr^{\perp}_{a}(b) = \vec{OB'} = \frac{\langle a,b\rangle}{|a|}$ $\boxed{\langle a,a\rangle = |a|^2}$

$Pr^{\perp}_{a}(b) = pr^{\perp}_{a}(b)\cdot\frac{a}{|a|} = \frac{\langle a,b\rangle}{\langle a,a\rangle}a$

$Pr^{\perp}_{b}(a) = pr^{\perp}_{b}(a)\cdot\frac{b}{|b|} = \frac{\langle b,a\rangle}{\langle b,b\rangle}b$

(vector) (scalar) (unit vector)

$\boxed{\cos\measuredangle(a,b) = \frac{Pr^{\perp}_{a}(b)}{|b|} = \frac{pr^{\perp}_{b}(a)}{|a|}}$

✓ ORTHONORMAL FRAMES.

$\mathcal{B}$ basis $= (e_1, \dots, e_m)$ of $\mathbb{V}^m$ is:

- orthogonal if $e_i \perp e_j$, $(\forall)\, i \neq j$
- orthonormal if $\begin{cases} e_i \perp e_j, (\forall)\, i \neq j \\ |e_i| = 1, \forall i \\ \text{(unit vectors)} \end{cases}$

✓ orthonormal / orthogonal basis ⇒ orthonormal / orthogonal frame.

✓ COORDINATES OF A POINT P w.r. orthonormal frame

$[P]_K = \begin{bmatrix} x_1 \\ \vdots \\ x_m \end{bmatrix} = \begin{bmatrix} |a|\cdot\cos\measuredangle(a,e_1) \\ \vdots \\ |a|\cdot\cos\measuredangle(a,e_m) \end{bmatrix} = |a|\cdot\begin{bmatrix} \cos\measuredangle(a,e_1) \\ \vdots \\ \cos\measuredangle(a,e_m) \end{bmatrix} = [\vec{OP}]_{\mathcal{B}}$.

$P(x_1, \dots, x_m)$

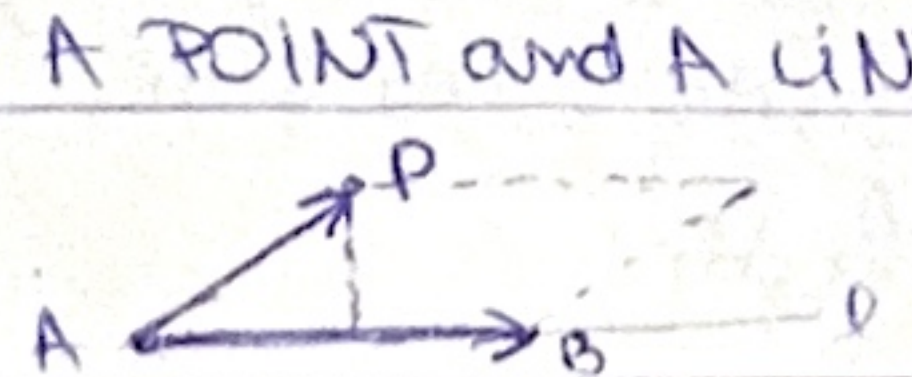
$a = \vec{OP}$ position vector.

✓ DISTANCE BETWEEN TWO POINTS $P, Q \in \mathbb{E}^m$:

$d(P,Q) = |\vec{QP}| = \sqrt{\langle\vec{QP},\vec{QP}\rangle} = \sqrt{(p_1-q_1)^2 + \dots + (p_n-q_n)^2}$

✓ DISTANCE BETWEEN A POINT and A LINE

$d(l,P) = \frac{|\vec{AP} \times v|}{|v|}$



✓ CROSS PRODUCT $(a \times b)$

- $a \parallel b \Rightarrow a \times b = 0$ (vector prod.)
- $a \nparallel b \Rightarrow$
  1) $|a \times b| = A_{parall.}$
  2) $a \times b \perp a$, $a \times b \perp b$
  3) $(a, b, a\times b)$ r-or basis of $\mathbb{V}^3$.

$v = v_{\parallel} + v_{\perp}$

? $\boxed{a \times b = |a|\cdot J^{\perp a}(Pr^{\perp a}(b))}$

? • billinear: ...

• skew-symm: $v \times w = -w \times v$

✓ POSITION VS. DIRECTION VECTOR

- POSITION vector: origin, $(\vec{OP})$ identifies a point's location
- DIRECTION vector: displacement, describes orientation, $(\vec{AB})$ independent of any fixed point in the space.

✓ ORIENTED ANGLES (plane $\mathbb{A}^2$, not space $\mathbb{A}^3$) → ∃ bij. between set of or. angles / unit circle

− $\measuredangle(h,k)$ = exterior of the angle (not between h and k)

$\measuredangle_{or}(h,k) = \begin{cases} \measuredangle(h,k) \text{ if } (\vec{OA},\vec{OB}) \text{ r-or.} \\ -\measuredangle(h,k) \text{ if } (\vec{OA},\vec{OB}) \text{ l-or.} \end{cases}$

→ set of or. angles + addition = ABELIAN GROUP

✓ SYMMETRIC VS SKEW-SYMM.

Symm: $A^T = A$

Skew-symm: $A^T = -A$

✓ SINE of an ORIENTED ANGLE

$\sin\measuredangle_{or}(a,b) = \frac{pr^{\perp}_{J(a)}(b)}{|b|} = -\frac{pr^{\perp}_{J(b)}(a)}{|a|}$

COSINE oriented / unoriented (same)

$\cos\measuredangle_{or}(a,b) = \frac{pr^{\perp}_{a}(b)}{|b|} = \frac{\langle a,b\rangle}{|a|\cdot|b|} \overset{?}{=} \frac{pr^{\perp}_{b}\, a}{|a|}$

$\sin\measuredangle_{or}(v,w) = \frac{[v,w]}{|v|\cdot|w|} = \frac{2\cdot \text{Area}_{or}(ABC)}{|AB|\cdot|AC|}$

✓ COSINE OF ANGLE BETWEEN 2 PLANES

$\cos\theta = \frac{|\langle n_1,n_2\rangle|}{|n_1|\cdot|n_2|}$

✓ J-operator

$J : \mathbb{V}^2 \to \mathbb{V}^2$ (unique) $\begin{cases} J(v) \perp v \; ; \; |J(v)| = |v|. \\ (v, J(v)) \text{ r-or. basis of } \mathbb{V}^2. \end{cases}$

✓ SCALAR PRODUCT = a real nr.

$\langle\_,\_\rangle : \mathbb{V} \times \mathbb{V} \to \mathbb{R}$; $\langle a,b\rangle = \begin{cases} 0, \text{ if } a = 0 \text{ or } b = 0 \\ |a|\cdot|b|\cdot\cos\measuredangle(a,b), \text{ oth.} \end{cases}$

1) Billinear: $\langle a\underline{v} + b\underline{w}, \underline{u}\rangle = a\langle\underline{v},\underline{u}\rangle + b\langle\underline{w},\underline{u}\rangle$

$\langle\underline{v}, a\underline{w} + b\underline{u}\rangle = a\langle\underline{v},\underline{w}\rangle + b\langle\underline{v},\underline{u}\rangle$

$\boxed{\langle(a,b),(-b,a)\rangle = 0}$

2) Symm: $\langle v,w\rangle = \langle w,v\rangle$

3) Pos-def: $v \neq 0 \Rightarrow \langle v,v\rangle > 0$.

4) $v \perp w \Leftrightarrow \langle v,w\rangle = 0$ (r-angle) ; $|v| = 1 \Leftrightarrow \langle v,v\rangle = 1$ (unit lengths)

✓ $\boxed{d(P,l) = \frac{|ax_0 + by_0 + c|}{\sqrt{a^2+b^2}}}$

✓ PROJECTION VECTOR

$\boxed{\vec{AC'} = \frac{\langle\vec{AC},v\rangle}{|v|^2}\cdot v}$

✓ SCALAR PROD ; orthonormal frame

$a(a_1, \dots, a_m)$, $b(b_1, \dots, b_m)$

$\boxed{\langle a,b\rangle = a_1b_1 + \dots + a_mb_m.}$ $\boxed{a \perp b \Leftrightarrow a_1b_1 + \dots + a_nb_n = 0}$

$\boxed{|a| = \sqrt{a_1^2 + a_2^2 + \dots + a_m^2}}$

$\cos\measuredangle(a,b) = \frac{a_1b_1 + \dots + a_mb_m}{\sqrt{a_1^2 + \dots + a_m^2}\cdot\sqrt{b_1^2 + \dots + b_m^2}} = \frac{\langle a,b\rangle}{|a|\cdot|b|}$

GEOMETRY REFERENCE SHEET.

$\mathbb{E}^2 = \mathbb{A}^m$ + scalar product

↓ euclidean space ↘ affine space (POINTS) ↓ measurement

$\mathbb{E}$ = affine space of dimension ≥ 3.

$\mathbb{E}^m$ = m-dimensional Euclidean space.

$\mathbb{V}^m$ = space of vectors in $\mathbb{E}^m$.

$\mathbb{A}^m$ = m-dimensional affine space.

ORIENTED SEGMENT

$[AB]$; $A \to B$.

FREE VECTOR

$B - A = \vec{AB}$

EQUIPOLENCY:

1) $[xy] \sim [AB]$ if:

$\begin{cases} [Ay], [Bx] \text{ have the same midpoints.} \\ [xy], [AB] \text{ oriented segments.} \end{cases}$

2) A, B, x, y non-collinear:

equipolency $\Leftrightarrow$ AByx parallelogram.

3) same magnitude + direction.

$\boxed{\vec{AB} : \text{the vector containing the oriented segment } [AB]}$

THE SET OF ALL VECTORS:

$\mathbb{V} = \{ \vec{AB} : A, B \in \mathbb{E} \} = \mathbb{E} \times \mathbb{E} / \sim$.

↓

(real) vector space ↗ vector addition ↘ scalar multiplication

LENGHT OF A VECTOR

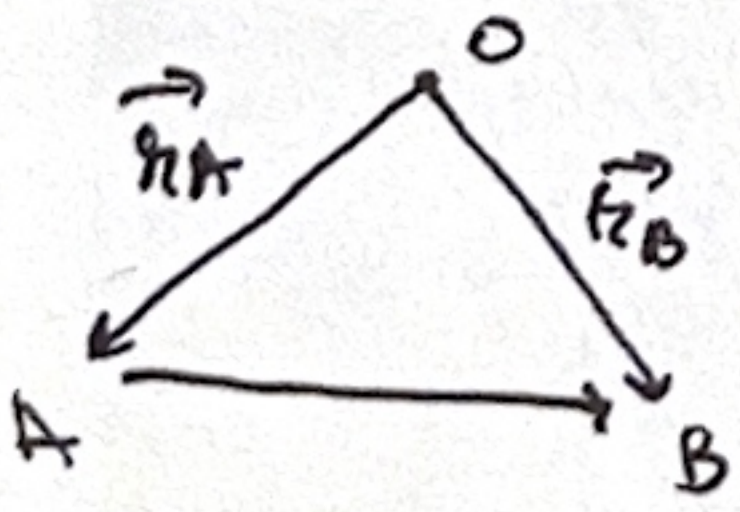
$|\vec{AB}| = |B - A|$.

$\vec{AA} = \vec{0}$.

THE POSITION VECTOR

Fix $O \in \mathbb{E}^m$.

$(\forall) A \in \mathbb{E}^m : \vec{r_A} = \vec{OA}$



$\vec{AB} = \vec{r_B} - \vec{r_A}$

- $A, M, B \in \mathbb{E}^m$

$\Rightarrow$ M midpoint of $[AB]$

$\Leftrightarrow$ $\vec{r_M} = \frac{1}{2}(\vec{r_A} + \vec{r_B})$.

VECTORS FORM A △

$\vec{u} + \vec{v} + \vec{w} = \vec{0}$.

$\boxed{\text{AFFINE SPACE + fixed origin} \Rightarrow \text{„AMBIENT" space}}$

AFFINE SPACE

set of points to which we attach $\mathbb{V}$ and 2 maps:

$\square - \square : \mathbb{A} \times \mathbb{A} \to \mathbb{V}$

$\square + \square : \mathbb{V} \times \mathbb{A} \to \mathbb{A}$

↳ 2 points $\Rightarrow$ vector.

↳ vector + point $\Rightarrow$ point.

$\xrightarrow{\text{axioms}}$

(AS₁): $(\forall) P$, $\vec{0} + P = P$.

(AS₂): $(\forall) P, (\forall) v, w$, we have $(v + w) + P = v + (w + P)$.

(AS₃): $(\forall) A, B$, we have $(A - B) + B = A$

(AS₄): $(\forall) P, (\forall) v$, we have $(v + P) - P = v$.

THEOREM

$K = (O, \mathcal{B})$

$K' = (O', \mathcal{B}')$

$(\forall) P \in \mathbb{A}^m$:

$[P]_{K'} = M_{B'B} \cdot ([P]_K - [O']_K)$.

$= M_{BB'}^{-1} ([P]_K - [O']_K)$

$= M_{B'B} \cdot [P]_K + [O]_{K'}$.

FRAME OF $\mathbb{A}^2$:

pair $K = (O, \mathcal{B})$, $\mathcal{B} = (i, j)$

CHANGING FRAMES:

The change of ~~matrix~~ basis matrix from $\mathcal{F}(f_1, \ldots, f_m)$ to $\mathcal{E}(e_1, \ldots, e_m)$ is

$$M_{\mathcal{E}\mathcal{F}} = \begin{array}{c} \left[\begin{array}{cccc} \vdots & \vdots & \cdots & \vdots \end{array} \right] \\ \begin{array}{cccc} \downarrow & \downarrow & & \downarrow \\ [f_1]_{\mathcal{E}} & [f_2]_{\mathcal{E}} & & [f_m]_{\mathcal{E}} \end{array} \end{array}$$

$$\boxed{[v]_{\mathcal{E}} = M_{\mathcal{E},\mathcal{F}} \cdot [v]_{\mathcal{F}}}$$

FRAMES ORIENTATION: (relative)

det ↗ $> 0 \Rightarrow$ same orientation ↘ $< 0 \Rightarrow$ opposite orientation.

DIRECTION SPACE

$D(\mathbb{A})$: $\mathbb{V}$ attached to $\mathbb{A}$.

$\dim(D(\mathbb{A})) = \dim(\mathbb{A})$.

AFFINE SUBSPACE

$\mathbb{A}$ affine space.

$S \subseteq \mathbb{A}$ is affine subspace

if: $\begin{cases} \exists P \\ \exists\, \mathbb{U} \leq D(\mathbb{A}) \text{ (vector subspace)} \end{cases}$ s.t.

$S = \mathbb{U} + P = \{ v + P, v \in \mathbb{U} \}$

↓

unique subspace $= D(S)$.

S is a POINT

$\dim \mathbb{U} = 0$.

S is a LINE

$\dim \mathbb{U} = 1$.

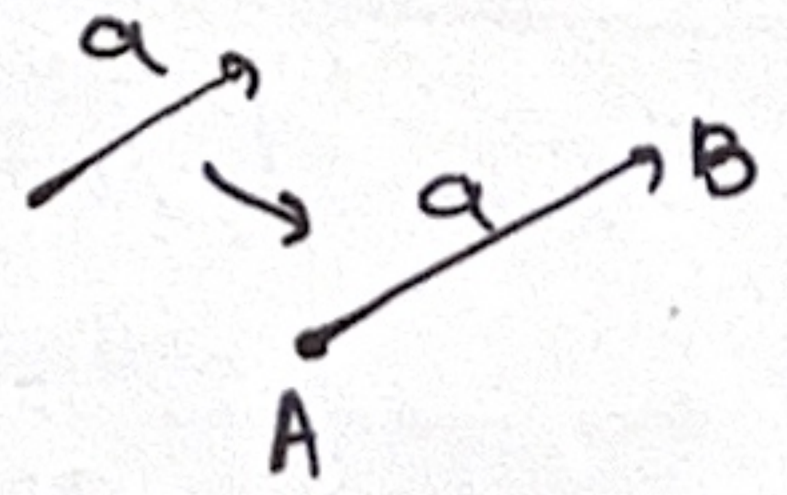
S is a PLANE

$\dim \mathbb{U} = 2$.

S is a hyperplane

$\dim \mathbb{U} = \dim \mathbb{A}^m - 1$.

„ACT ON A POINT"



vector a, point A $\Rightarrow \exists!\, B$ s.t. $a = \vec{AB}$.

POLYGON:

$\vec{A_1A_2} + \vec{A_2A_3} + \dots + \vec{A_{m-1}A_m} = \vec{0}$.

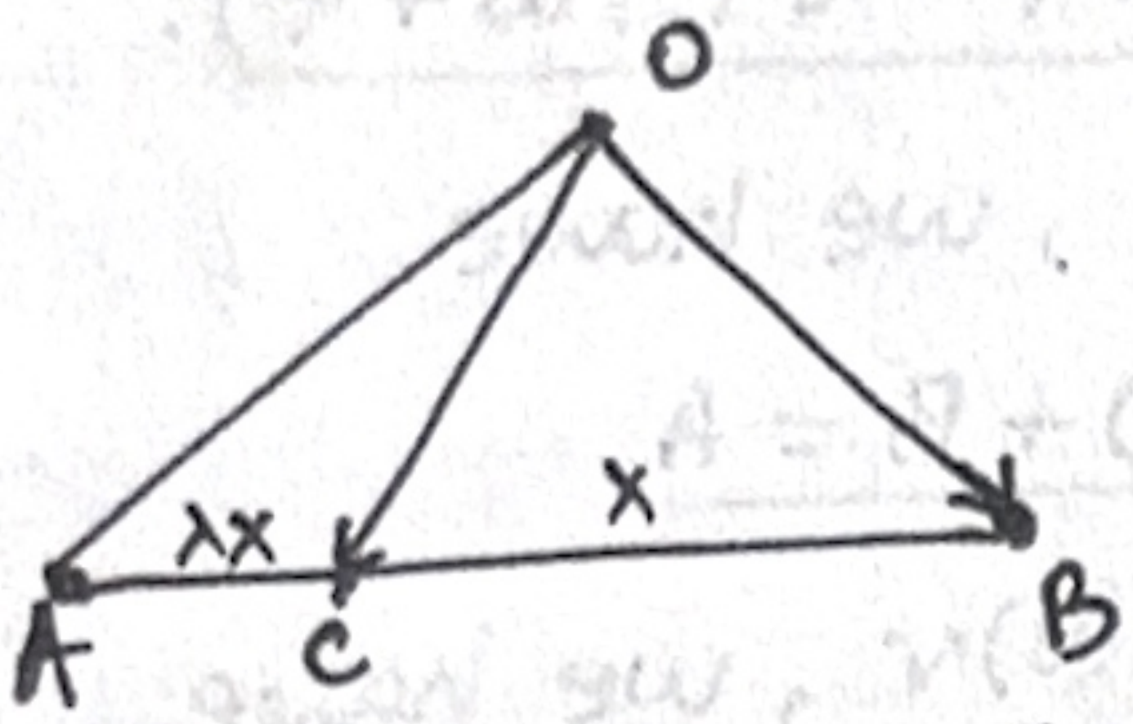
(all these points $\in \mathbb{E}^m$).

Δ inequality

$\|u\| + \|v\| > \|u+v\|$.

Lin. indep:

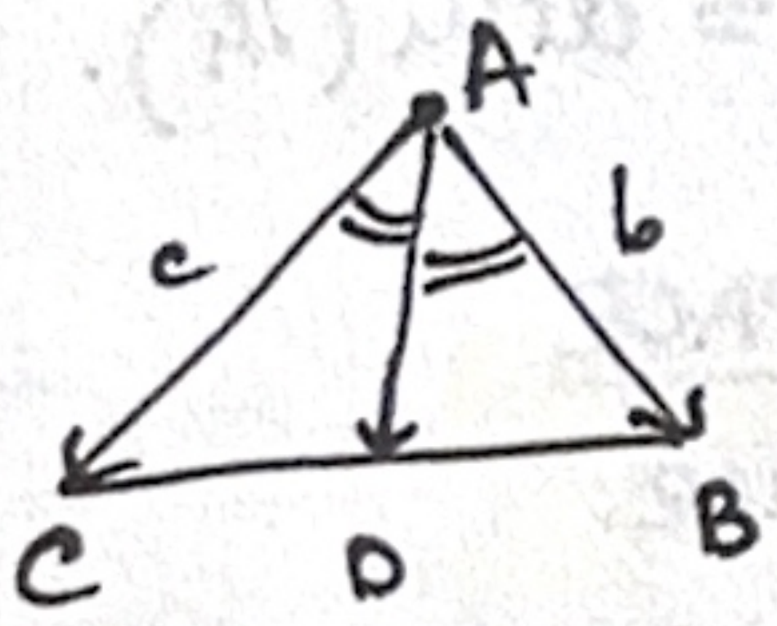
- det $\neq 0$
- visually



$$\vec{OC} = \frac{1}{1+\lambda}\vec{OA} + \frac{\lambda}{1+\lambda}\vec{OB}.$$

Bisector theorem:

$$\frac{AC}{AB} = \frac{CD}{DB}$$



COLLINEARITY

1) $m_{PQ} = mQR$

$\Rightarrow$ P, Q, R collinear.

2) $\left.\begin{array}{l}\vec{PQ} = \vec{r}_Q - \vec{r}_P \\ \vec{QR} = \vec{r}_R - \vec{r}_Q\end{array}\right\}$

$\Rightarrow$ lin dep. $\Rightarrow$ P, Q, R collinear.

TEMPLATES.

1) $a=?$ s.t. (u,v,w) same or. as $\mathcal{B}$?

$M=\begin{bmatrix} \underset{u}{\downarrow} & \underset{v}{\downarrow} & \underset{w}{\downarrow} \end{bmatrix} \to \det M \overset{?}{} 0$ → $>0 \Rightarrow$ same or.

↳ $<0 \Rightarrow \neq$ or.

↳ $=0 \Rightarrow$ lin dep.

$\Rightarrow$ no basis.

2) K': $(B, \vec{AB}, \vec{AC}, \vec{AD})$.

$\vec{BP} = x\vec{AB} + y\vec{AC} + z\vec{AD}$

$\vec{BA} = \dots \Rightarrow [A]_{K'}$

$\vec{BC} = \dots \Rightarrow [C]_{K'}$

$\vec{BD} = \dots \Rightarrow [D]_{K'}$

$M=\begin{bmatrix} \underset{\vec{AB}}{\downarrow} & \underset{\vec{AC}}{\downarrow} & \underset{\vec{AD}}{\downarrow} \end{bmatrix} \dots$

3) setup; $[B]_{K'} = ?$

$[B]_{K'} = M_{B'B} \cdot ([B]_K - [O']_K)$.

$M_{BB'} = \begin{bmatrix} \underset{i'}{\downarrow} & \underset{j'}{\downarrow} \end{bmatrix} \Rightarrow M_{BB'}^{-1} = \dots$ (gauss) J.)

4) hyperplane = dim − 1.

- plane in $\mathbb{A}^3$: $3-1=2 \to$ hyp. ✓
- point in $\mathbb{A}^1$: $1-1=0 \to$ hyp. ✓
- line in $\mathbb{A}^2$: $2-1=1 \to$ hyp. ✓
- ~~plane~~ ($\mathbb{A}^n$ with dim $= n-1$).

5) line containing 2 points:

$M(2,4)$, $N(2,-5)$

dir. vector: $\vec{MN} = (0,-9) = v$

$$\begin{cases} x = x_M + t\cdot v_x = 2 + t\cdot 0 \\ y = y_M + t\cdot v_y = 4 + t\cdot(-9) \end{cases}$$

6) plane containing A, axis Ox.

$A(1,5,7)$, $O(0,0,0)$; $(1,0,0)$

~~dir.~~ vector: $\vec{OA} = (1,5,7)$

$$\begin{cases} x = 0 + 1\cdot s + 1\cdot t \\ y = 0 + 0\cdot s + 5\cdot t \\ z = 0 + 0\cdot s + 7\cdot t \end{cases}$$

$O(0,0,0)$ · dir. vector $(1,0,0)$ · another vector.

$(0,-7,5)\cdot(1,5,7) = 0$

$(0,-7,5)\cdot(1,0,0) = 0$

$\Rightarrow n = (0,-7,5)$ normal vector.

π: $0\cdot(x-x_0) - 7\cdot(y-y_0) + 5(z-z_0) = 0$.

7) line contained in plane

$$l \in \pi \Leftrightarrow \begin{cases} v\cdot n = 0 \\ P \in H. \end{cases}$$

8) relative pos. of lines:

$$l_1 = \begin{cases} x = -3t \\ y = 2+3t \\ z = 1 \end{cases}, \quad l_2: \begin{cases} x = 1+5s \\ y = 1+13s \\ z = 1+10s \end{cases}$$

$\Downarrow$ $(-3,3,0)$ dir. v. $\Downarrow$ $(5,13,10)$ dir. v.

$\Rightarrow$ NOT PROPORTIONAL $\Rightarrow l_1 \nparallel l_2$.

solve system $\Rightarrow$ sol $\Rightarrow \cap$.

9) collinear, coplanar?

- not prop. $\Rightarrow$ not $\parallel$ $\Rightarrow$ not coll.
- coplanar $\Leftrightarrow V_{\text{parallelip}} = 0$.

$\Leftrightarrow |\langle a\times b, c\rangle| = [a,b,c] = 0$

$\Leftrightarrow \left|\begin{vmatrix} \vec{AB} \\ \vec{AC} \\ \vec{AD} \end{vmatrix}\right| = 0$

10) eq. plane containing BCD.

$K = (B, \vec{AB}, \vec{AC}, \vec{AD})$.

$\vec{BP} = x\cdot\vec{AB} + y\cdot\vec{AC} + z\cdot\vec{AD}$

$\vec{BC} = \dots \Rightarrow [C]_K$

$\vec{BD} = \dots \Rightarrow [D]_K$

$[B]_K = (0,0,0)$.

$n = \vec{BC}\times\vec{BD} = (\_,\_,\_)$

(BCD): $\_(x-x_B) + \_(y-y_B) + \_(z-z_B)$.

11) $K(O,\mathcal{B}) \to l$

$K'(O',\mathcal{B}') \to l'$.

$M_{B'B} \to M_{BB'}^{-1} \dots$

$\begin{bmatrix} x \\ y \end{bmatrix} = M_{BB'}^{-1} \cdot \begin{bmatrix} x' \\ y' \end{bmatrix} + [O']_K$.

12) orthogonal proj. of C on AB.

$\vec{AP} = \Pr^{\perp}_{AB}(AC) = \dfrac{\langle AC, AB\rangle}{\langle AB, AB\rangle}\cdot AB$

↓ (direction of projection)

$P = A + \vec{AP}$ = projection point.

13) orthogonal proj. of $\vec{CB}$ on $\vec{CA}$

$\vec{v} = \Pr^{\perp}_{CA}(CB) = \dfrac{\langle CB, CA\rangle}{\langle CA, CA\rangle}\, CA \dots$

14) line passing through M with n.

$M(-\frac{1}{2}, \frac{1}{2})$, $n = (-3,1)$.

$-3(x-x_M) + 1\cdot(y-y_M) = 0$.

$\Rightarrow y = 3x+2$.

15) $\cos\sphericalangle(p,q) = \dfrac{\langle p,q\rangle}{|p|\cdot|q|}$

15) dist. between planes:

$$\begin{cases} \pi_1: x - 2y - 2z + 7 = 0 \\ \pi_2: 2x - 4y - 4z + 17 = 0. \end{cases}$$

- $n_1 (1,-2,-2)$
- $n_2 (2,-4,-4)$

$\Rightarrow$ proportional $\Rightarrow \pi_1 \parallel \pi_2$.

$\Rightarrow d = \dfrac{|d_1 - d_2|}{\sqrt{a^2+b^2+c^2}} \dots$

16) angle between planes.

$\pi_1 \Rightarrow n_1(1,-\sqrt{2},1)$

$\pi_2 \Rightarrow n_2(1,\sqrt{2},-1)$

$\cos\theta = \dfrac{|\langle n_1,n_2\rangle|}{|n_1|\cdot|n_2|} \dots$

$\langle n_1,n_2\rangle = 1-2-1 = -2$.

$|n_1| = \dots$

$|n_2| = \dots$

DISTANCES

✓

1) from a point to a hyperplane; a_{n-1}

$$d(Q,S) = \frac{|a_1 q_1 + \dots + a_n q_n - b|}{\sqrt{a_1^2 + \dots + a_n^2}}$$

✓

2) from a point to a line/plane.

$$d(P,l) = \frac{|a\cdot x_P + b\cdot y_P + c\cdot z_P + d|}{\sqrt{a^2+b^2+c^2}}$$

✓

3) between to $\parallel$ planes:

$$d(\pi_1,\pi_2) = \frac{|d_1-d_2|}{\sqrt{a^2+b^2+c^2}}.$$

✓

4) between two skew lines:

$$d(l_1,l_2) = \frac{|\langle AC, u\times v\rangle|}{|u\times v|}$$

$A\in l_1$; u

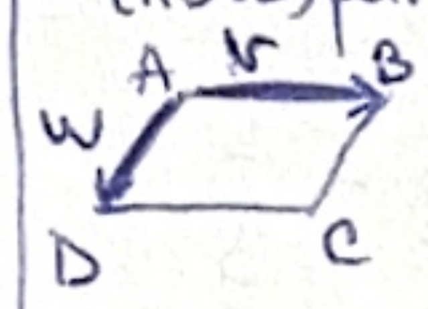
✓ $C \in l_2$; v.

5) from a point to a line

$$d(P,l) = \frac{|\vec{QP}\times v|}{|v|}$$

AREAS, VOLUMES

$A_{(ABCD)\,par.} = |AB\times AD| = b\cdot h = 2\cdot A_{\Delta ABC}$



$= \left|\begin{vmatrix} x_A & y_A & 1 \\ x_B & y_B & 1 \\ x_D & y_D & 1 \end{vmatrix}\right| = \left|\begin{vmatrix} v_x & v_y \\ w_x & w_y \end{vmatrix}\right|$

$A_{(ABCD)\,or.\,par} = [v\times w] = |v|\cdot|w|\sin\theta$

17) $v = ?$ s.t. $v\perp a$, $v\perp b$.

$\sphericalangle(v,Ox)$ acute; $|v| = 26$.

$u = a\times b = \dots$

acute angle with the x-axis $\Rightarrow x > 0$

$A_{\Delta ABC} = \frac{1}{2}\cdot\|\vec{AB}\times\vec{AC}\| = \frac{1}{2}\cdot\left|\begin{vmatrix} x_A & y_A & 1 \\ x_B & y_B & 1 \\ x_C & y_C & 1 \end{vmatrix}\right| =$

$= \frac{1}{2}|[\vec{AB},\vec{AC}]| = \frac{b\cdot h}{2}$

$= A_{or}(ABCD)/2. = \frac{1}{2}\cdot|\vec{AB}|\cdot|\vec{AC}|\cdot\sin\theta$

$V_{(ABCD)\,tetrah} = \frac{a\cdot h}{3} = \frac{1}{6}\cdot|[\vec{AB},\vec{AC},\vec{AD}]| =$

$= \frac{1}{6}\cdot|[u,v,w]| = \frac{1}{6}|\langle u\times v, w\rangle| = \frac{1}{6}\cdot\left|\begin{vmatrix} x_A & y_A & z_A & 1 \\ \vdots \end{vmatrix}\right|$

$= \frac{1}{6}\cdot\left|\begin{vmatrix} x_B-x_A & y_B-y_A & z_B-z_A \end{vmatrix}\right| \dots$

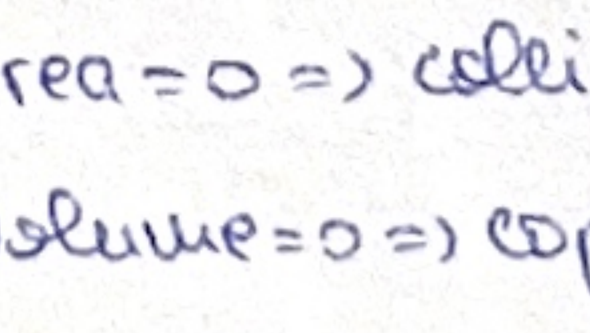
2D: $[a,b] = 0 \Rightarrow$ Area $= 0 \Rightarrow$ collinear.

3D: $[a,b,c] = 0 \Rightarrow$ Volume $= 0 \Rightarrow$ coplanar.

18) direction of common $\perp$: $n = v_1\times v_2$.

19) orthogonal proj. of C on AB.; v = dir. vector $(\vec{AB})$

$\vec{AC'} = \dfrac{\langle\vec{AC}, v\rangle}{|v|^2}\cdot v$



$C' = A + \vec{AC'}$ (A is on the plane already).

20) orthogonal proj. of D on (ABC); $\vec{AB} = \dots$, $\vec{AC} = \dots$

$n = \vec{AB}\times\vec{AC} = (2,0,1)$; $A(-1,-1,1)$

(ABC): $2(x-x_A) + 0\cdot(y-y_A) + 1\cdot(z-z_A) = 0$.

$h[\ v\ ]\ v_H$

$\vec{AD'} = \dfrac{\langle\vec{AD}, n\rangle}{|n|^2}\cdot n$

✓✓ CROSS PRODUCT $(a\times b)$✓ ✓POSITION VS. DIRECTION VECTOR

PARALLEL PROJECTION ON A HYPERPLANE

K of $\mathbb{A}^2$. Project onto $l: x+y-1=0$ in the direction of $v(-2,-1)$.

$P(x_P, y_P)$

$l_P: \begin{cases} x = x_P - 2t \\ y = y_P - t \end{cases}$ (line passing through P in direction of v).

$P' = l \cap l_P$ = projection of P onto l.

plug $l_P \to l$:

$\Rightarrow x_P - 2t + y_P - t - 1 = 0$

$\Rightarrow x_P + y_P - 3t - 1 = 0.$

$\Rightarrow t = \frac{1}{3}(x_P + y_P - 1).$

$$\Rightarrow [P']_K = \begin{bmatrix} \frac{1}{3}x_P - \frac{2}{3}y_P + \frac{2}{3} \\ -\frac{1}{3}x_P + \frac{2}{3}y_P + \frac{1}{3} \end{bmatrix} = \frac{1}{3}\cdot\begin{bmatrix} 1 & -2 \\ -1 & 2 \end{bmatrix}\begin{bmatrix} x_P \\ y_P \end{bmatrix} + \frac{1}{3}\begin{bmatrix} 2 \\ 1 \end{bmatrix}.$$

$x = x_P - 2\cdot\frac{1}{3}x_P - \frac{2}{3}\cdot y_P + \frac{2}{3}$

$x = \frac{1}{3}x_P - \frac{2}{3}y_P + \frac{2}{3}$

$y = y_P - 2\cdot\frac{1}{3}x_P - 2\cdot\frac{1}{3}y_P + \frac{2}{3}$

$y = -\frac{1}{3}x_P + \frac{2}{3}y_P + \frac{1}{3}$

PROJECTION of THE POINT P onto H parallel to v.

K; $\mathbb{A}^m$; $H = a_1x_1 + \dots + a_mx_m + a_{m+1} = 0$.

l_P passes through $P(p_1, \dots, p_m)$

$l_P = \{P + t\cdot v : t \in \mathbb{R}\}$

~~$l_P \cap H$~~ $P + tv \in l \cap H$.

$$P' = P - \frac{a_1p_1 + \dots + a_mp_m + a_{m+1}}{a_1v_1 + \dots + a_mv_m}\cdot v$$

PARALLEL REFLECTION in a hyperplane

$l: x+y-1=0.$

$v(-2,-1).$

$P(x_P, y_P).$

$l_P: \begin{cases} x = x_P - 2t \\ y = y_P - t. \end{cases}$

$$\boxed{P' = 2\,\mathrm{Pr}_{l,v}(P) - P}$$

$$\mathrm{Pr}_{H,v}(P) = \frac{P+P'}{2}$$

ORTHOGONAL REFLECTION

$\mathrm{Ref}^{\perp}_{H} = \mathrm{Ref}_{H,v}.$

PARALLEL PROJECTION ON A LINE

$l:$

$v(-2,-1)$

$l: \begin{cases} x = 1 - t \\ y = t \end{cases}$

$D(l_P): x - 2y = 0.$ dir. space.

$l_P: (x - x_P) - 2\cdot(y - y_P) = 0.$

$\Rightarrow (1 - t - x_P) - 2\cdot(t - y_P) = 0$

$\Rightarrow t = -\frac{1}{3}(x_P - 2y_P - 1).$

loci < constant distance
equidistant

- point → 2D: circle
  ↳ 3D: sphere : $(x-x_0)^2 + (y-y_0)^2 + (z-z_0)^2 = c^2$
- line → cylinder : $x^2 + y^2 = c^2$
- plane → two ∥ planes: